MATHS MANTHAN
• TUITION CLASSES •
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Subject: Mathematics | **CBSE CLASS 10 – TEST (SET – A)** | **Time:** 60 Minutes
Class: Class X (CBSE) | **Chapter 9: Some Applications of Trigonometry** | **MM: 25**

Student Name: _____ **Roll No.:** _____ **Date:** _____

GENERAL INSTRUCTIONS:

1. This question paper contains 11 compulsory questions divided into four Sections: A, B, C, and D.
2. **Section A** comprises 5 Multiple Choice Questions (MCQs) of 1 mark each (Q1 to Q5).
3. **Section B** comprises 2 Very Short Answer (VSA) questions of 2 marks each (Q6 and Q7).
4. **Section C** comprises 3 Short Answer (SA) questions of 3 marks each (Q8 to Q10).
5. **Section D** comprises 1 Long Answer (LA) question of 5 marks (Q11). An internal choice is provided.
6. Draw neat, labeled geometric figures wherever applicable. Use of calculators is strictly prohibited.

SECTION A: Multiple Choice Questions (5 × 1 = 5 Marks)

Q1. At some time of the day, the length of the shadow of a tower is equal to its height. Then, the Sun's altitude at that time is: [1]

(A) 30° (B) 45° (C) 60° (D) 90°

Q2. If a vertical pole of length 7.5 m casts a shadow 5 m long on the ground and at the same time, a tower casts a shadow 24 m long, then the height of the tower is: [1]

(A) 20 m (B) 40 m (C) 60 m (D) 36 m

Q3. From a point on the ground, which is 30 m away from the foot of a vertical tower, the angle of elevation of the top of the tower is found to be 60°. The height (in metres) of the tower is: [1]

(A) $10\sqrt{3}$ m (B) $30\sqrt{3}$ m (C) 60 m (D) 30 m

Q4. The length of the string of a kite flying 50 m above the ground with an angle of elevation of 60° is: [1]

(A) $\frac{100}{\sqrt{3}}$ m (B) $100\sqrt{3}$ m (C) 150 m (D) $\frac{50}{\sqrt{3}}$ m

Q5. A ladder 14 m long leans against a wall. If the foot of the ladder is 7 m from the wall, then the angle of elevation of the top of the wall is: [1]

(A) 30° (B) 45° (C) 60° (D) 90°

SECTION B: Very Short Answer Questions (2 × 2 = 4 Marks)

Q6. The length of the shadow of a tower on the plane ground is $\sqrt{3}$ times the height of the tower. Find the angle of elevation of the Sun. [2]

Q7. Find the length of the shadow on the ground of a pole of height 18 m when the angle of elevation θ of the Sun is such that $\tan\theta = \frac{6}{7}$. [2]

SECTION C: Short Answer Questions (3 × 3 = 9 Marks)

Q8. The angle of elevation of the top of a building from the foot of a tower is 30° and the angle of elevation of the top of the tower from the foot of the building is 60°. If the tower is 50 m high, find the height of the building. **[3]**

Q9. From a point on a bridge across a river, the angles of depression of the banks on opposite sides of the river are 30° and 45° respectively. If the bridge is at a height of 4 m from the banks, find the width of the river. (Use $\sqrt{3} = 1.732$) **[3]**

Q10. The shadow of a tower standing on level ground is found to be 40 m longer when the Sun's altitude is 30° than when it is 60°. Find the height of the tower. **[3]**

SECTION D: Long Answer Question (1 × 5 = 5 Marks)

Q11. A pole 6 m high is fixed on the top of a tower. The angle of elevation of the top of the pole observed from a point P on the ground is 60° and the angle of depression of the point P from the top of the tower is 45°. Find the height of the tower and the distance of point P from the foot of the tower. (Use $\sqrt{3} = 1.73$) **[5]**

— OR —

From the top of a building 60 m high, the angles of depression of the top and bottom of a vertical lamp post are observed to be 30° and 60° respectively.

(i) Find the horizontal distance between the building and the lamp post.

(ii) Find the height of the lamp post and the distance between the tops of the building and the lamp post.

[5]

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★★★ END OF QUESTION PAPER · ALL THE BEST ★★★